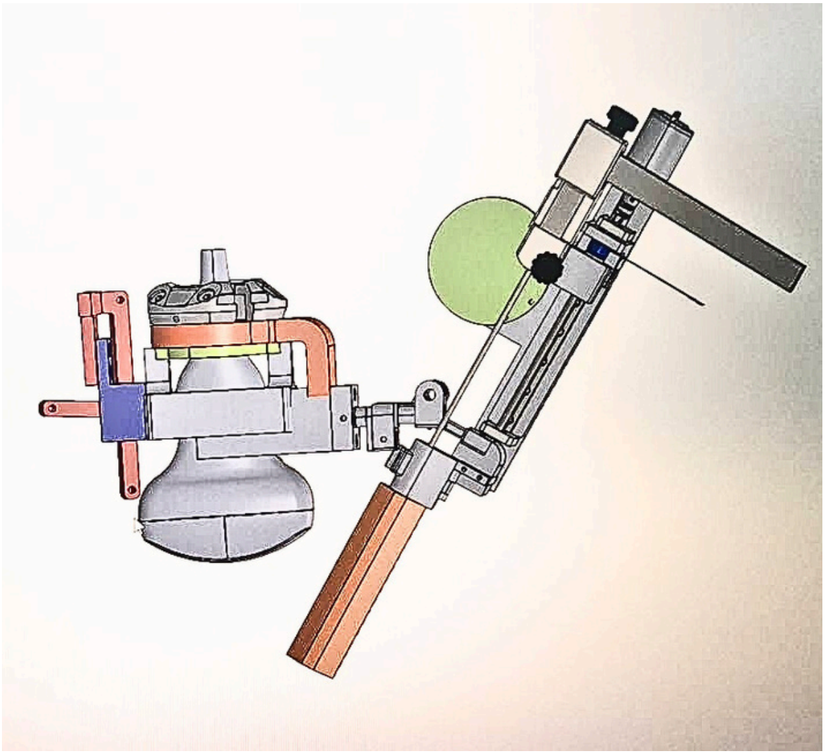
Needle Insertion Mechanism

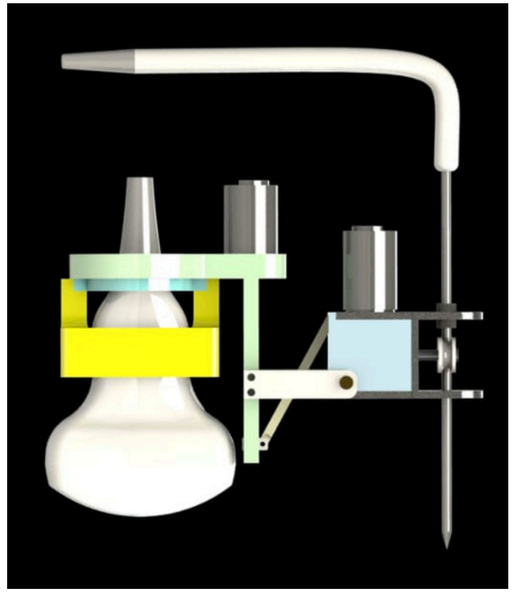
Shobit Verma

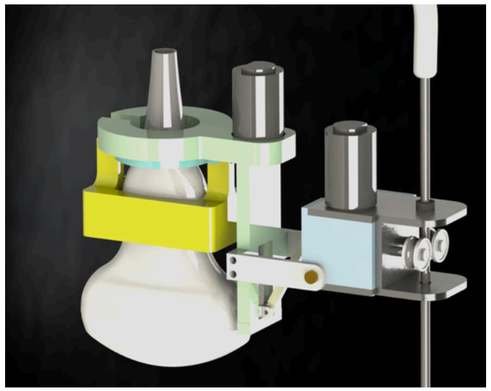
DESIGN PPT

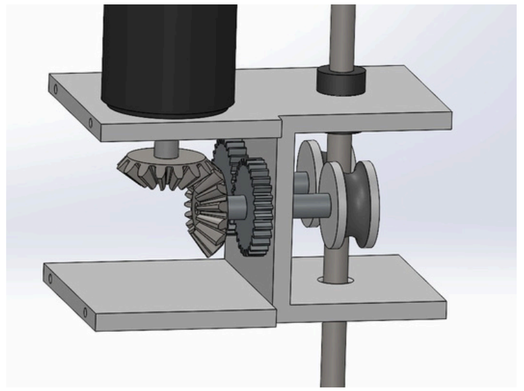
Existing problems of the original model.

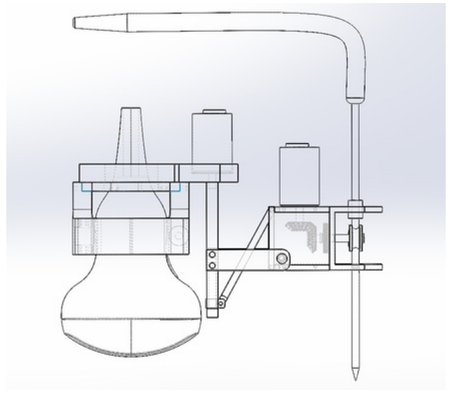
- Too bulky
- Cannot change the angle of inclination of needle during operation
- missing on haptic feedback bcoz of the lack of compliance in the system

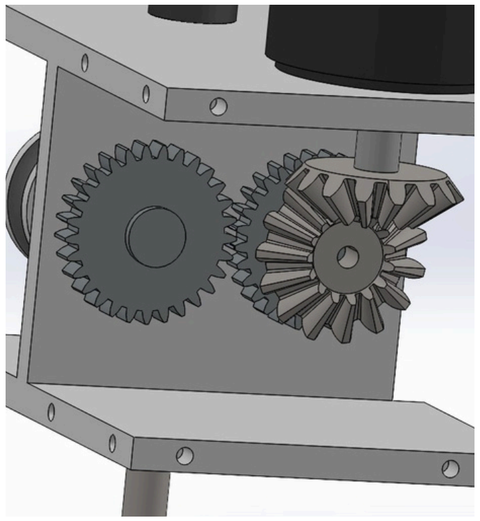


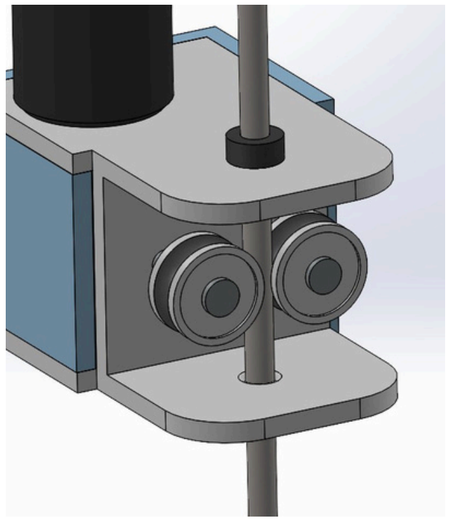






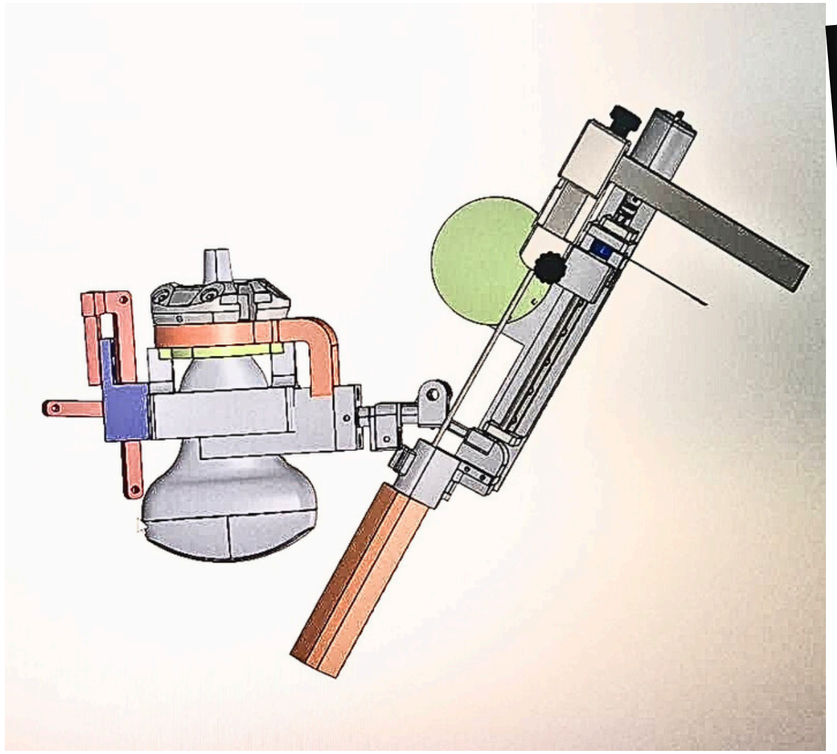




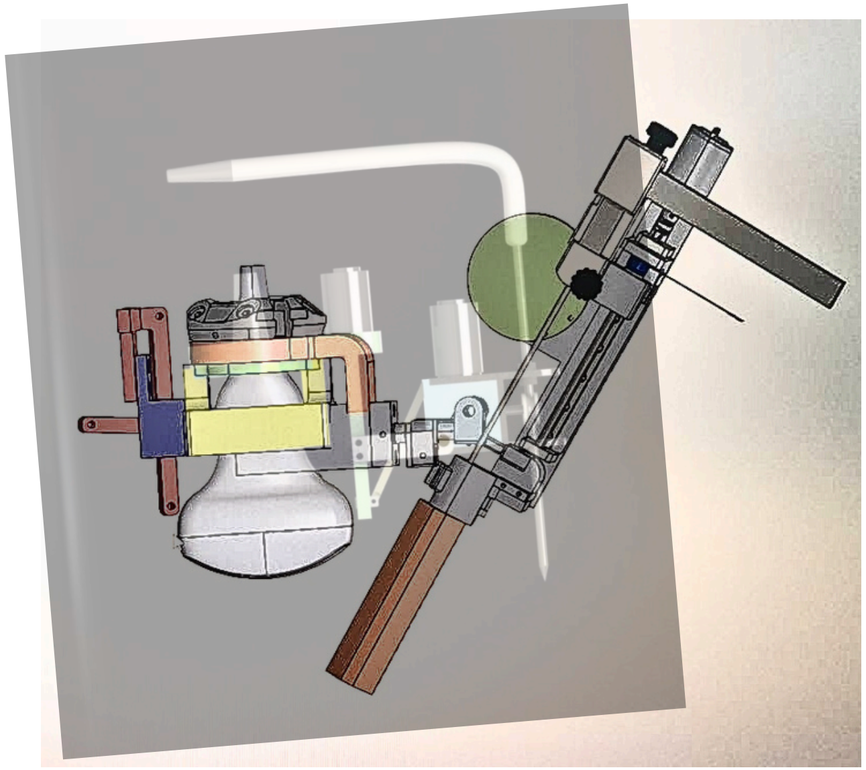


DESIGN -1 (CONCEPT)

Size Comparison

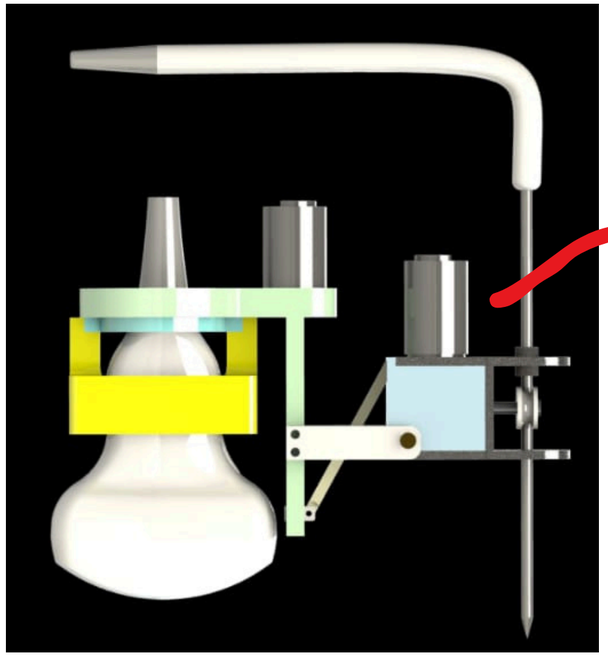


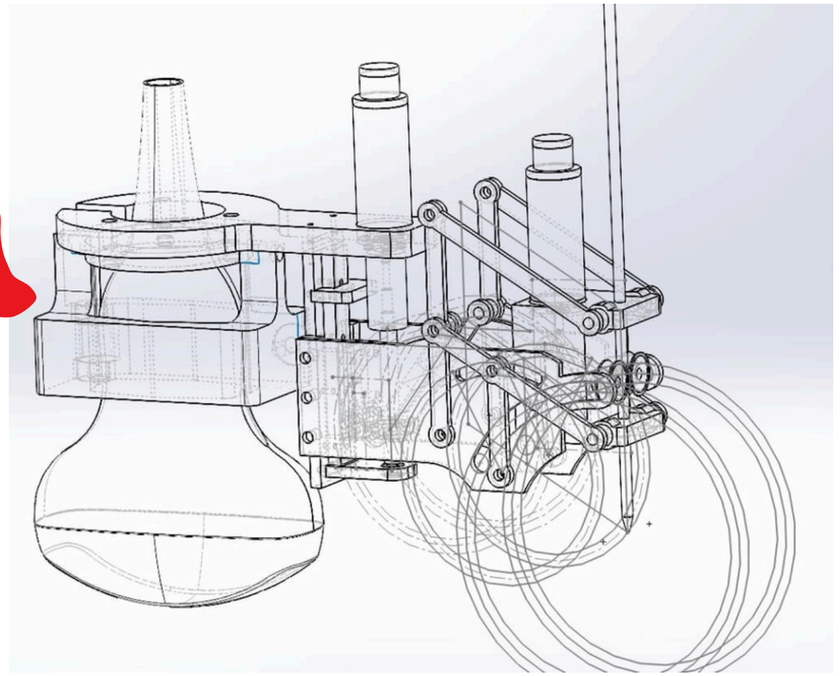


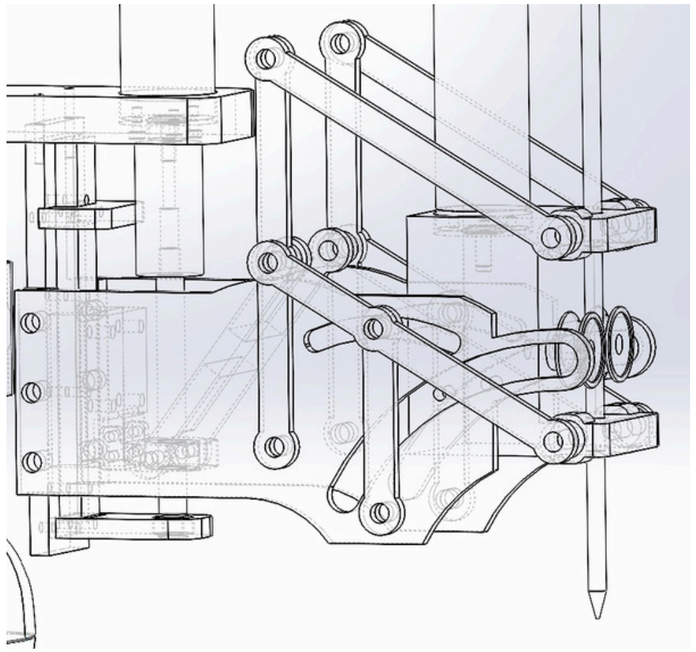


2 DOF FREEDOM ACHIEVED, NEEDLE DISPENSION MADE COMPLIANT, COMPACT SIZE

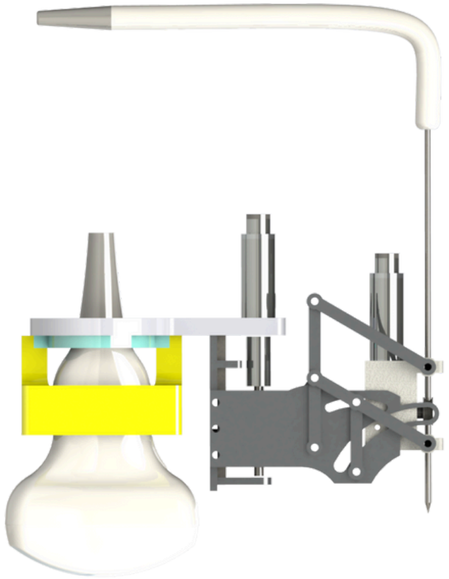
Designed the first working version.

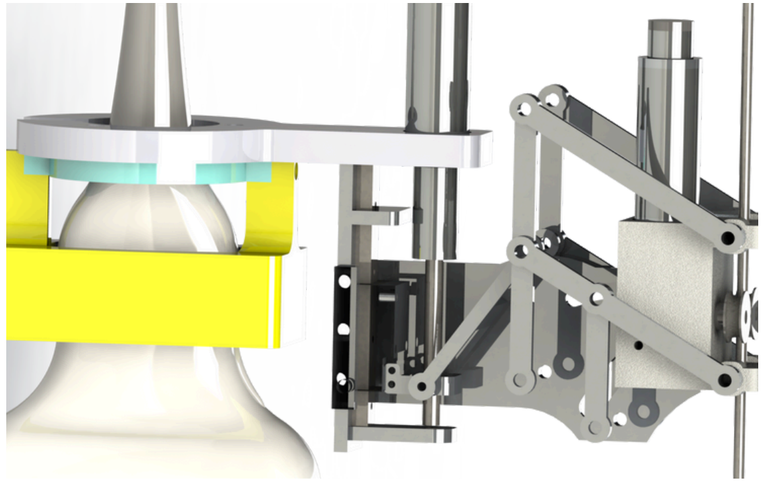


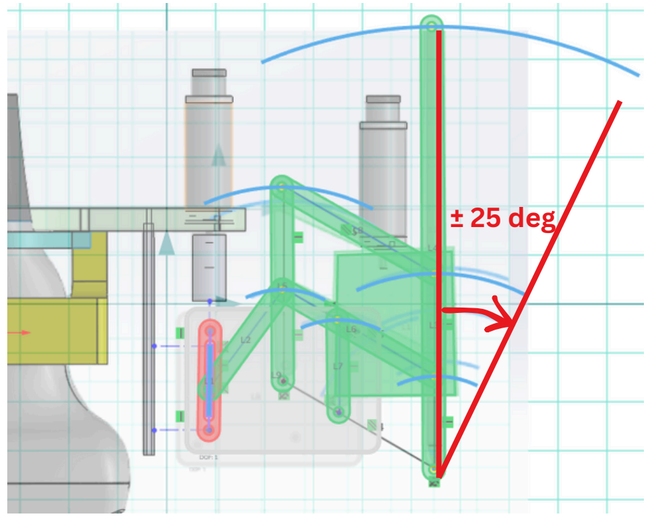


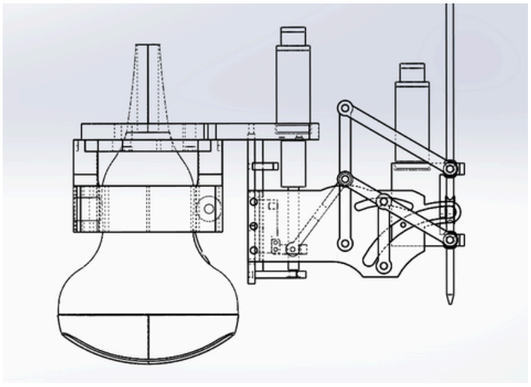


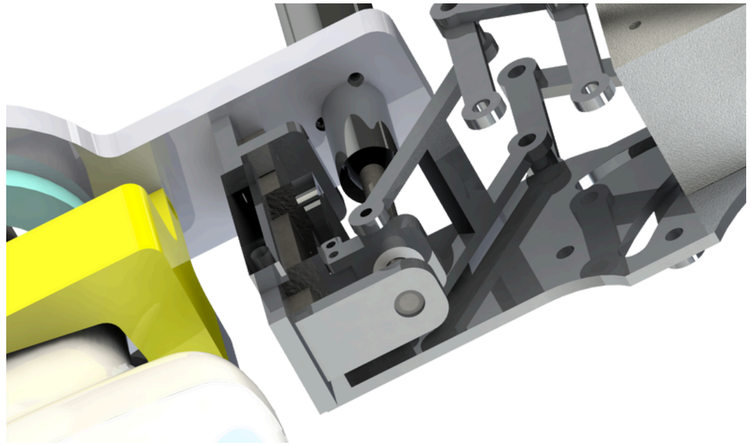
With Remote center of motion (RCM) at the needle end

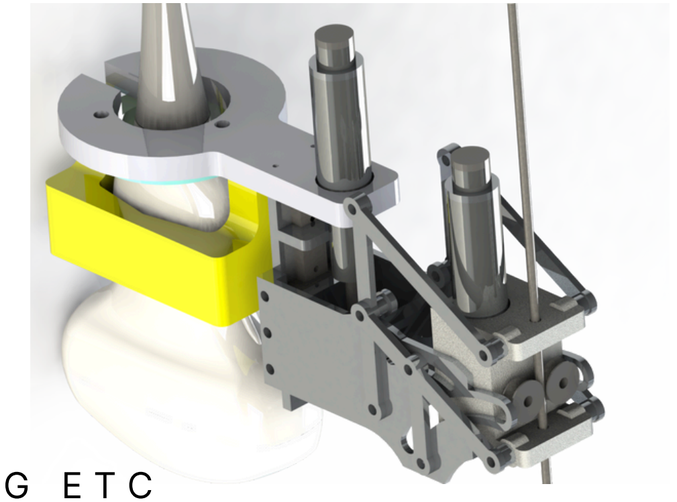






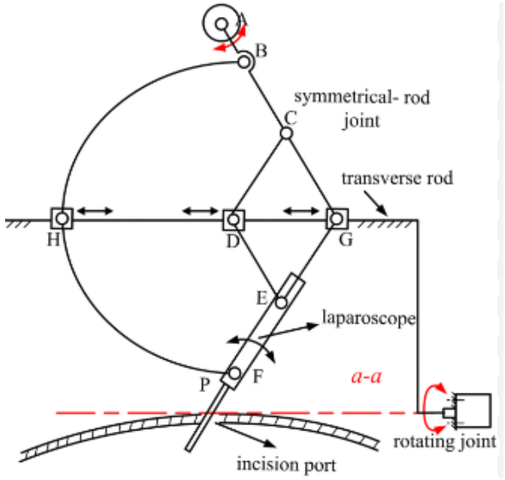


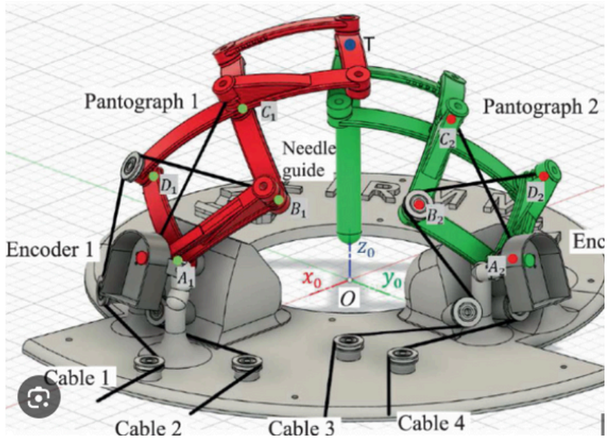


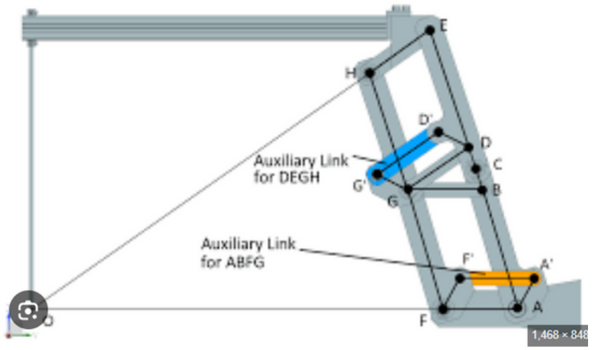


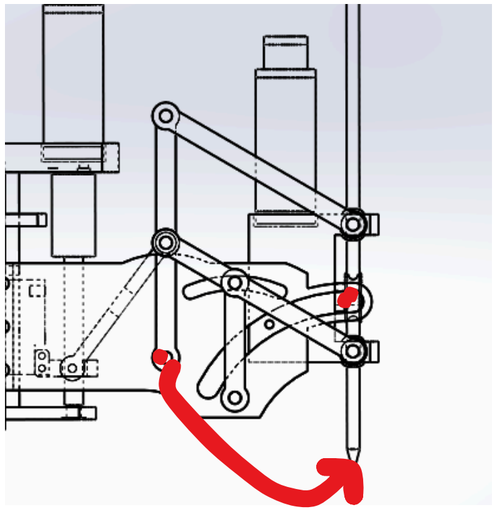
CAREFULLY ADDED ALL THE BEARINGS, JOINTS, COUPLING ETC

RCM (Remote center of motion) implementation



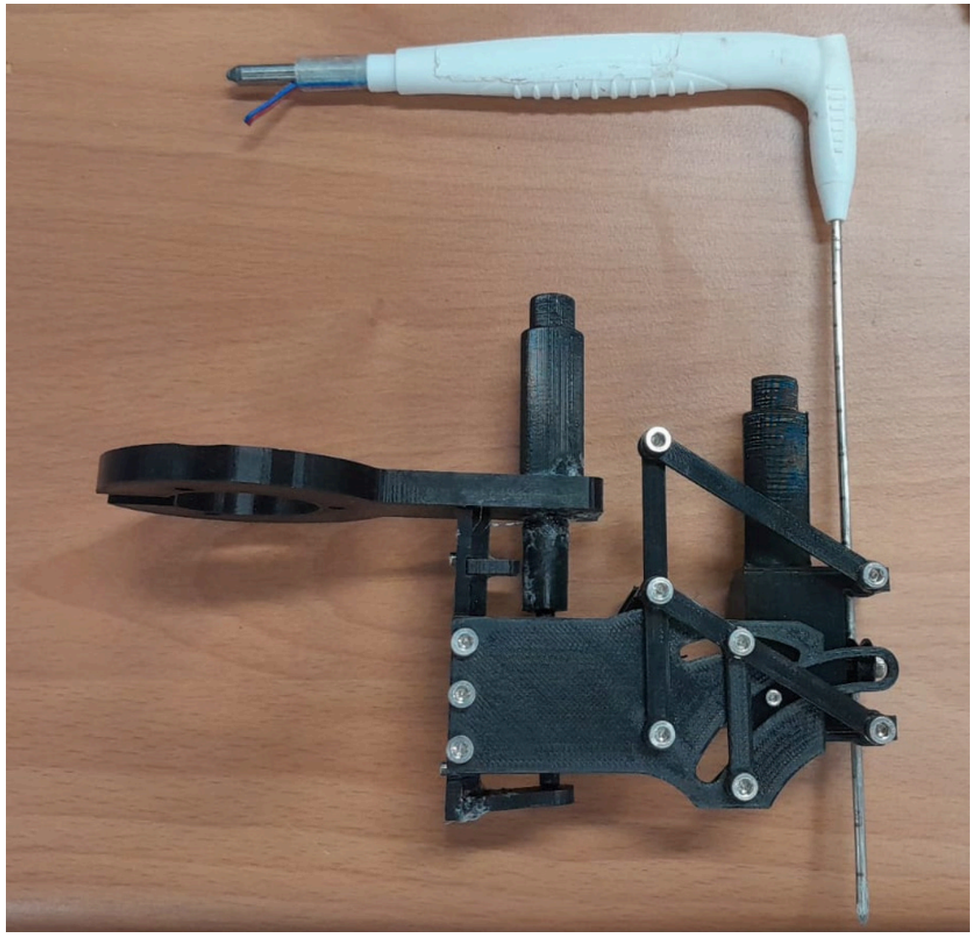


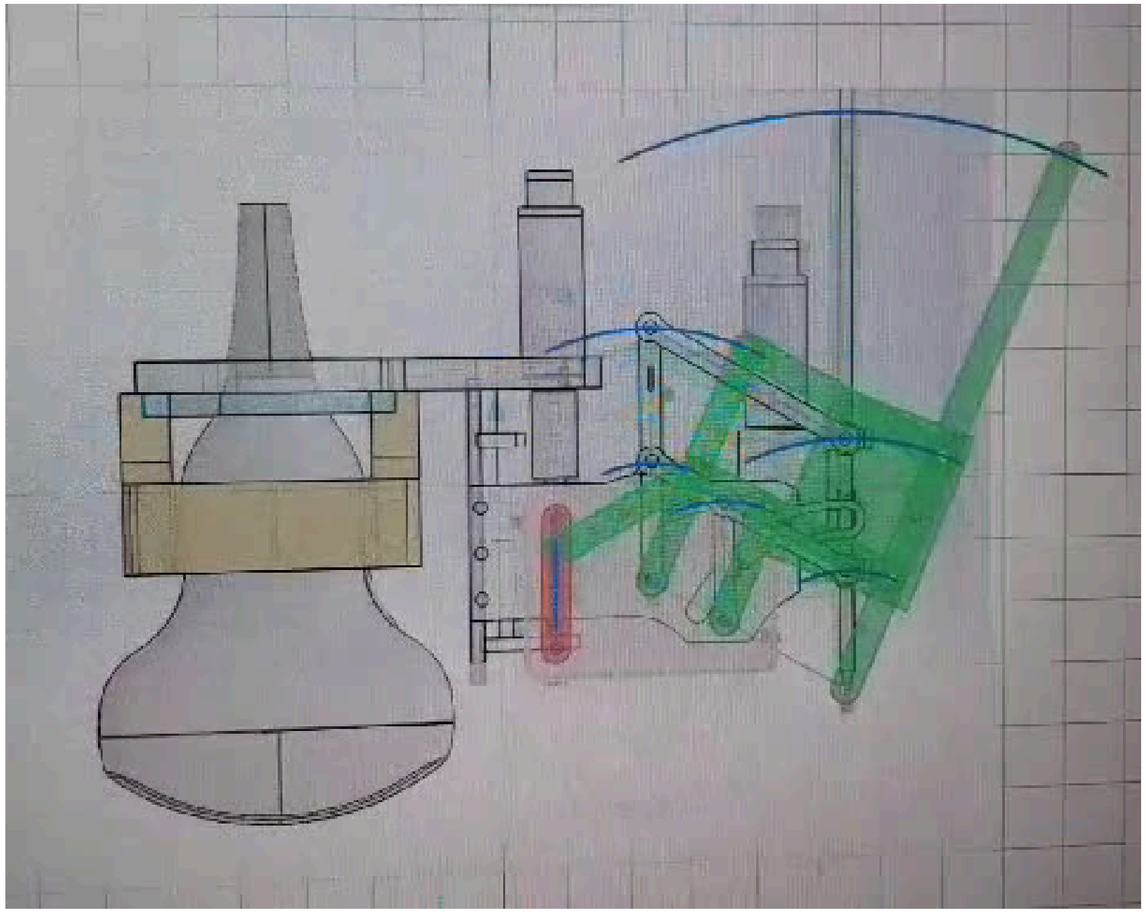




A remote fixed point, with no physical revolute joint over there, around which a mechanism or part of it can rotate

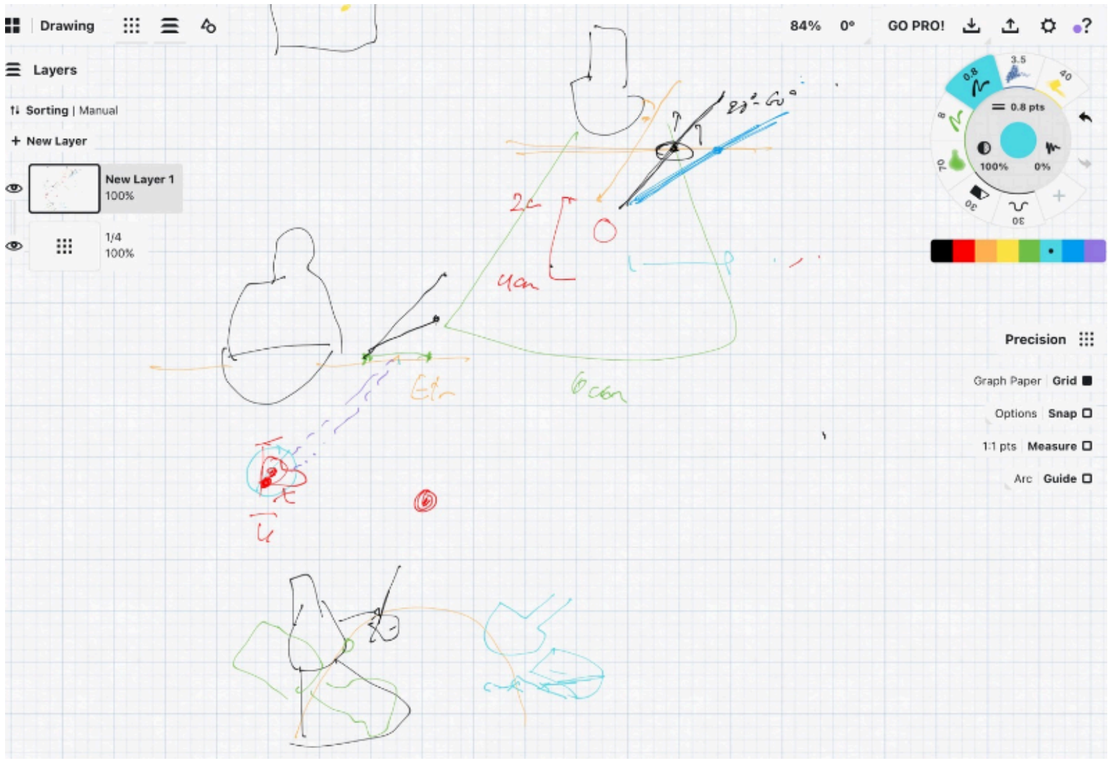
Assembled the model-

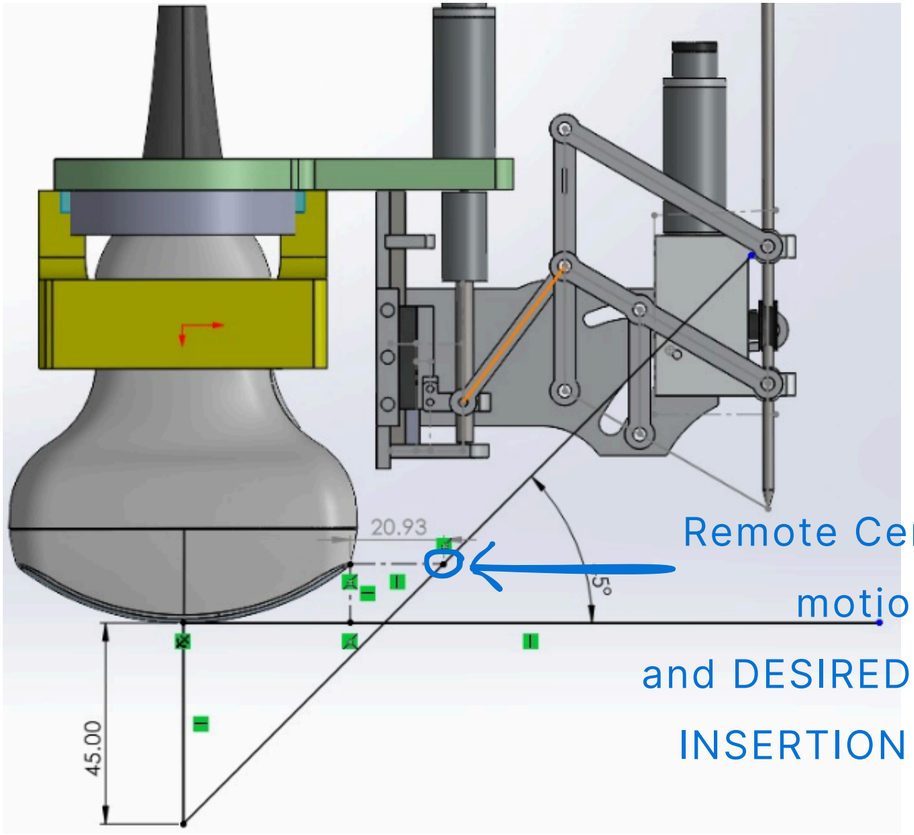


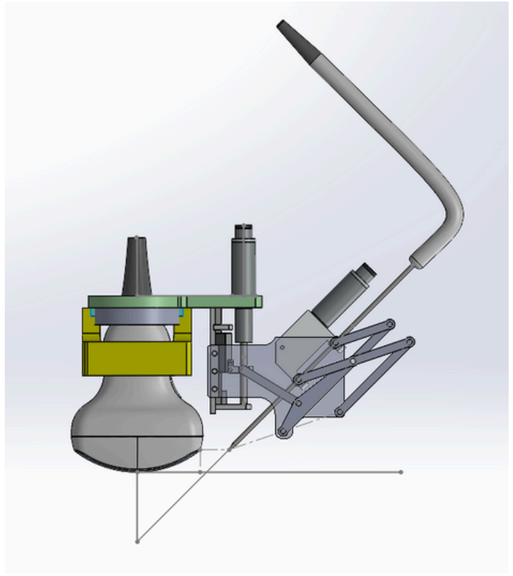


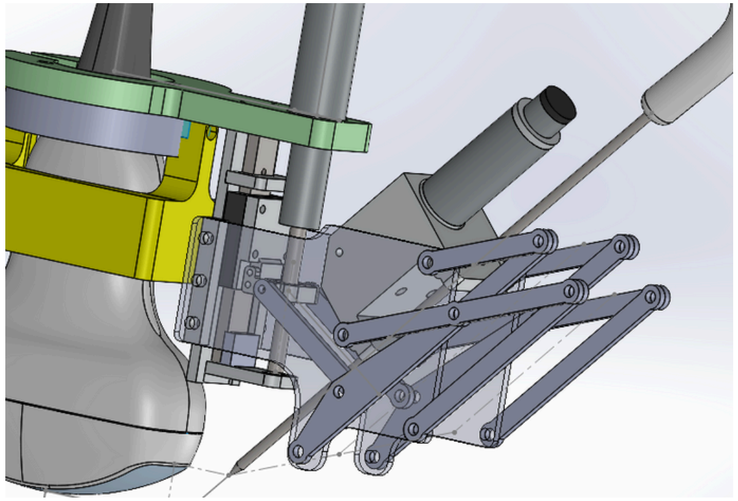
Needed to change the entry point and position of RCM according to surgery requirements

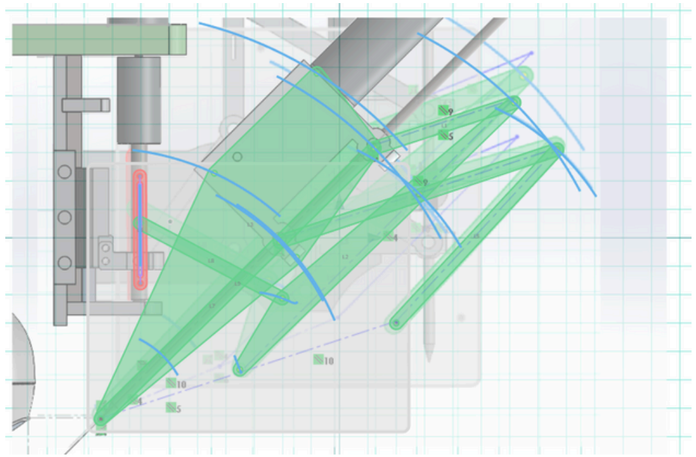
Needed to shift the RCM at needle insertion point

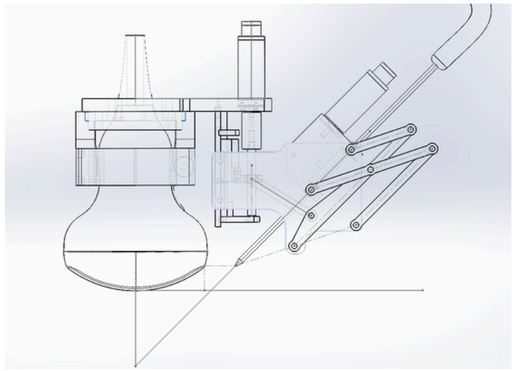


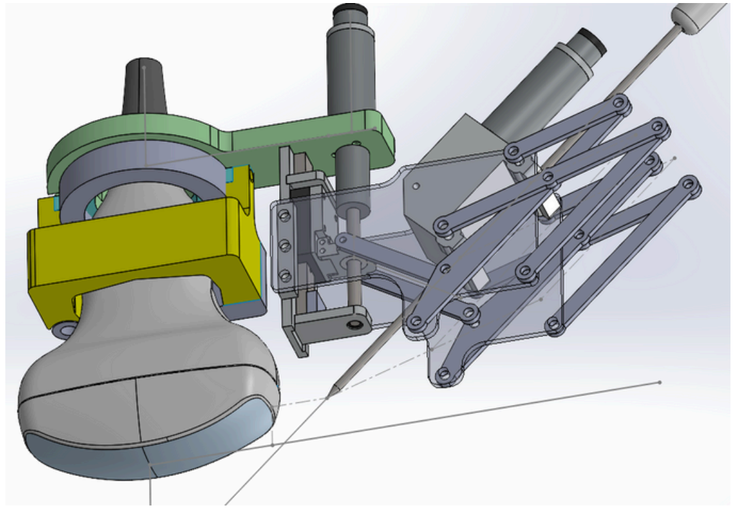


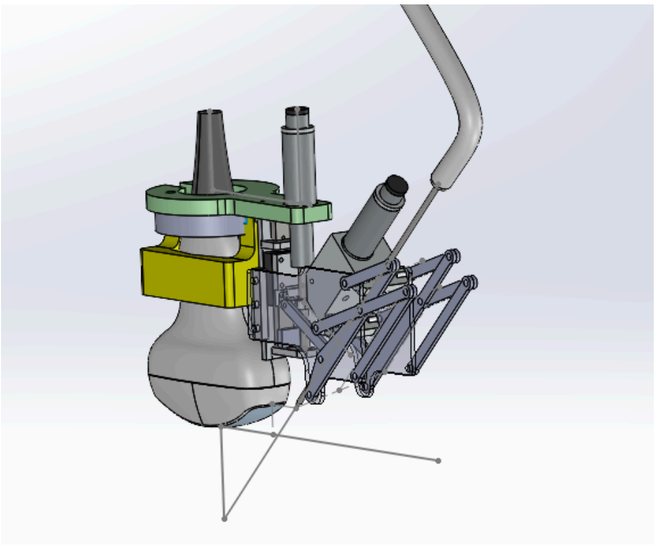








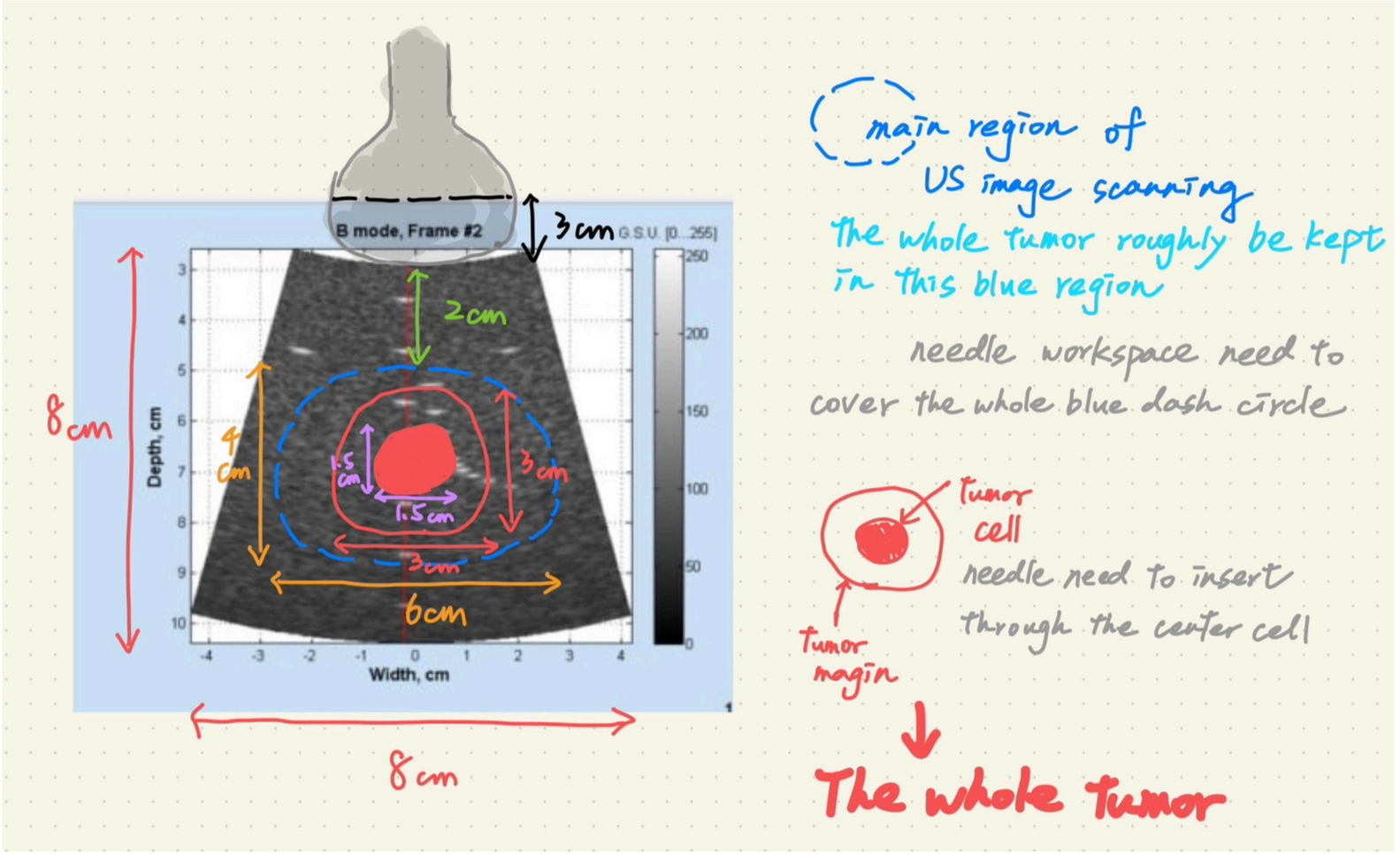




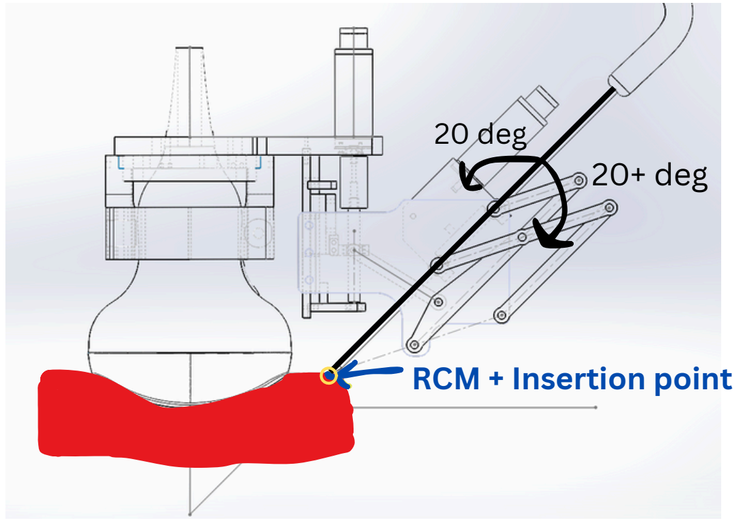
THE NEW MODEL

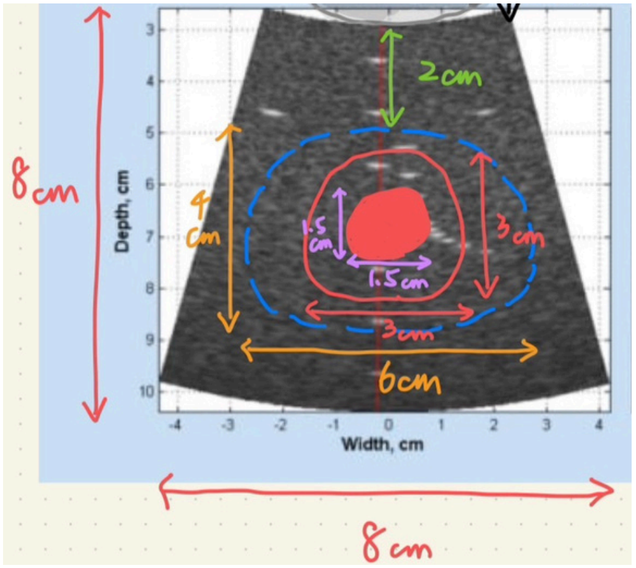
Workspace analysis

- After the last week problem, we started the analysis from the root level again
- Tina did the literature survey to find out the exact range of motion we need
- Figured out a way how to reach all the area mentioned by keeping the RCM fixed at the insertion point.

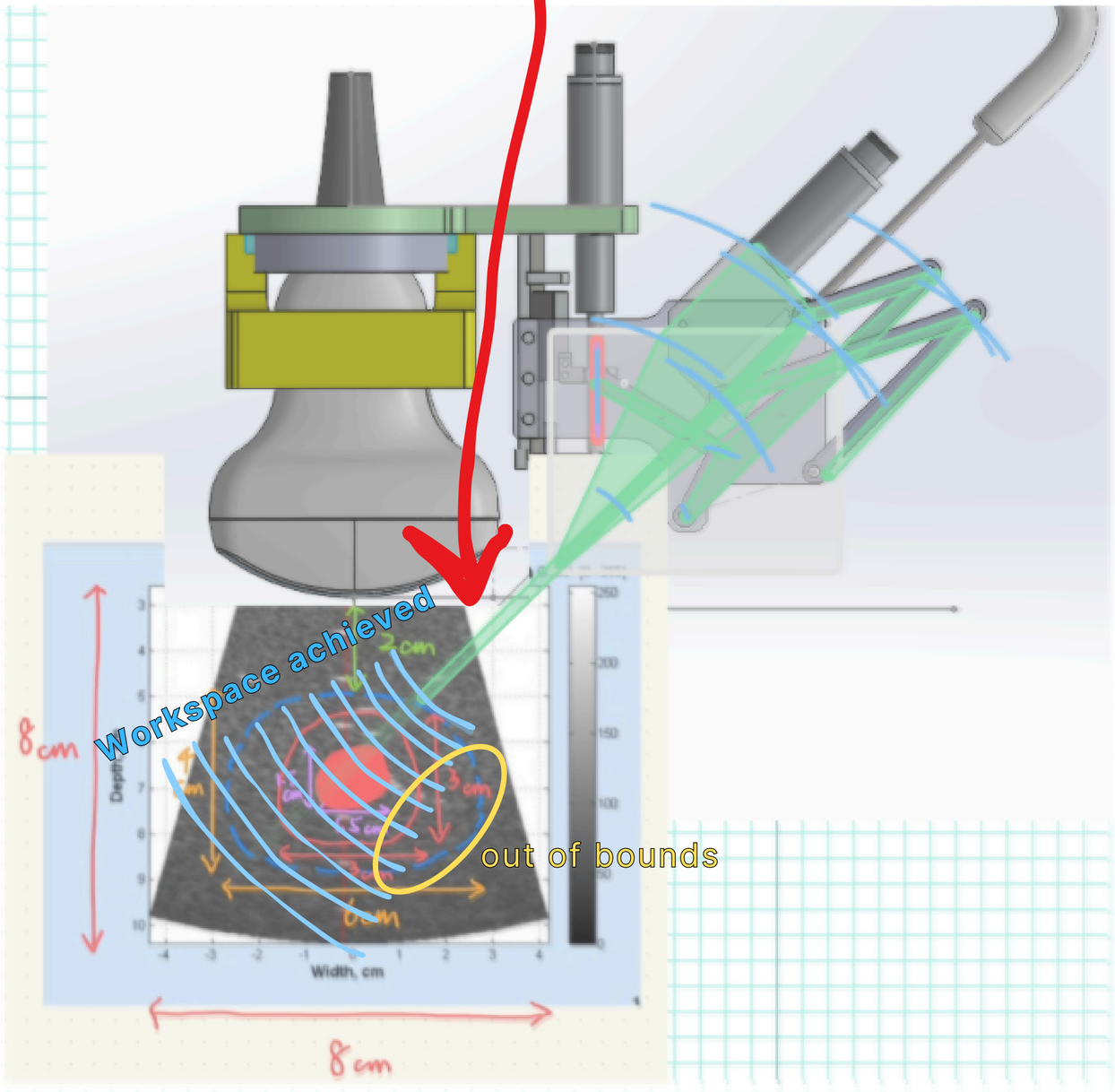


Workspace achieved by latest design

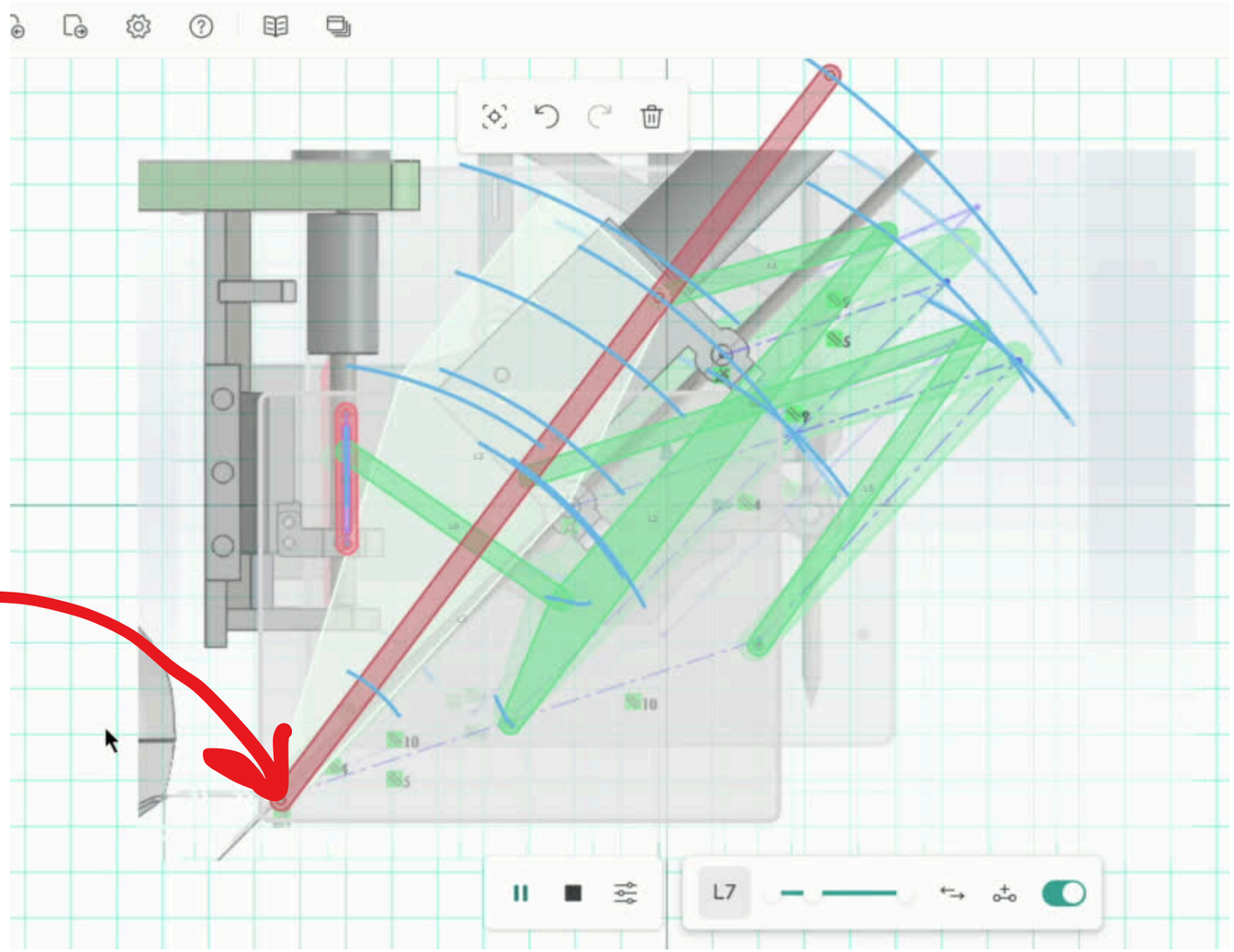




Required workspace



RCM
&
needle insertion
point



MOTION SIMULATION